Joint cell–gene–protein embeddings for interpreting drug responses: a stress test of SCENE

- **Paper:** Møberg et al., *Setting the SCENE for Interpretable Cell–Gene Embeddings in Single-Cell RNA-seq*, bioRxiv (2026) — https://www.biorxiv.org/content/10.64898/2026.09.12.750699v1
- **Code:** https://github.com/oscarmoeberg/SCENE (`pip install scene-ldm`) · reproducibility repo: https://github.com/oscarmoeberg/SCENE-reproducibility
- **Data:** all public (GEO / scvi-tools / pertpy loaders), no access requests needed.

What SCENE does, intuitively

SCENE places every **cell and every gene as points in the same small space** (even 2D or 3D). A gene is likely to be expressed in a cell, and at a higher level, if the two points are close. This works from raw counts and uses all genes, with no normalization and no "top-2000 variable genes" filtering. So you can read biology straight off the map: marker genes sit next to their cell types, and genes that respond together should cluster together.

Our project in one line

We extend SCENE from one modality to two (**RNA + surface proteins**) and use it to ask whether **drug responses and drug targets** form readable structure in the embedding. We also check honestly how much of that structure is real biology and how much is technical artifact.

Fit with the course (CBIO7215, Multimodal AI for therapeutic discovery)

Course requirement	How we address it
Multimodal	RNA + protein (CITE-seq) in one shared embedding with modality-specific likelihoods. The SCENE authors list this as untested future work.
Therapeutic-discovery problem	Drug-response interpretation: do the genes that respond to a drug, and its known targets (Open Targets / DrugBank), move together in the embedding? Can the embedding recognize a drug's mechanism of action (MoA)?
Rigor over performance; negative results OK	Pre-registered hypotheses, a fair expression-matched null test, an audit for technical artifacts, several seeds, and baselines.
Public / approved data	Datasets listed below. We will confirm them with the instructor before starting.

Datasets (all public)

Dataset	Modality	Why
PBMC CITE-seq (Hao et al. 2021)	RNA + 224 proteins	Main multimodal dataset; 8 donors → leave-one-donor-out tests
NeurIPS 2021 bone marrow CITE-seq	RNA + 134 proteins	Second multimodal dataset; differentiation lineages
IFN-β stimulation (Kang et al. 2018)	RNA	Interferon-β is an approved drug; response program is well characterized
Glucocorticoid time course (dexamethasone)	RNA	Drug response over time (1–18 h), used in the SCENE paper
sci-Plex (Srivatsan et al. 2020), subset	RNA	Cancer cell lines treated with 188 compounds with annotated MoA

Timeline

- **Saturday:** set everything up, write the code, smoke-test it, and launch the training queues on both laptops in the evening.
- **Weekdays (~2 days of compute):** runs go unattended. Check the log once a day.
- **Next weekend:** analyse the results, combine the two tracks, and write up.

Who does what

	Person A: Multimodal model (ML track)	Person B: Drug response and validation (biology + stats track)
Big idea	Make SCENE **multimodal**. Add proteins as a third type of point, with their own count model (NB). Also test better count models for RNA and a way to embed new patients (donors) without retraining.	Test whether the embedding is **useful and trustworthy for therapeutic questions**. Check whether drug-response genes and drug targets form coherent groups, and whether that holds up against a fair null test and technical confounders.
Sat, together (~1.5 h)	Shared repo and environment; download datasets; time one training run; write down hypotheses *before* running.	Same session.
Sat, task 1	**Protein modality (~2.5 h):** add ADT counts as extra nodes with an NB likelihood and a per-protein background term. Cells, genes and proteins then share one space.	**Baseline embeddings (~1.5 h):** train default RNA-only SCENE on IFN-β, dexamethasone and a sci-Plex subset (3 seeds). Compute baselines: PCA loadings, LDVAE (linear scVI), GenePT.
Sat, task 2	**RNA likelihood variants (~1.5 h):** Negative Binomial and a "decoupled" variant (separate parameters for "detected?" vs. "how much?").	**Drug gene sets (~1.5 h):** IFN-β and GR response genes, targets of the sci-Plex compounds from Open Targets / DrugBank, MoA classes. Technical control sets: ribosomal, mitochondrial, cell cycle, dissociation stress, high-dropout genes.
Sat, task 3	**New donors (~1.5 h):** "fold-in" (freeze genes and proteins, fit only the new cells) vs. a zero-training baseline: cell = expression-weighted average of its gene positions.	**Fair null test (~2 h):** the paper compares modules against *uniformly random* genes. We compare against random genes *matched on expression level and detection rate*.
Sat, task 4	**Experiment configs (~1 h):** {RNA-only, RNA+protein} × {ZIP, NB, decoupled} × 2 dims (3D, 16D) × 2 CITE-seq datasets × 3 seeds, plus leave-one-donor-out on PBMC vs. totalVI.	**Artifact audit (~1.5 h):** how much of a gene's position is explained by technical covariates (mean expression, dropout, variance)? Measured with R^2 and kNN purity.

	Person A: Multimodal model (ML track)	Person B: Drug response and validation (biology + stats track)
Sat evening, together (~1 h)	Smoke test: 5 epochs on a small subset → launch the queue (`caffeinate`, laptop plugged in).	Same; launch own queue.
Weekdays	Check the log once a day (~5 min).	Same. Tasks 3–4 can also be finished here; they are light computations on top of the saved embeddings.
Next Sat (~4–5 h)	Tables and plots. Does protein help separate similar cell types? Does each protein land next to its own gene (CD8 protein ↔ *CD8A*)? Held-out likelihood, integration metrics, label-transfer accuracy for new donors.	**MoA test:** compute each drug's "shift" in gene space (treated vs. control) and check whether drugs with the same MoA shift similarly (sci-Plex). Run the null test and artifact audit on all embeddings, including Person A's variants.
Next Sun, together (~4 h)	**Joint question:** does adding protein, or a better likelihood, make drug-response structure clearer and less technical? Write Methods and Results (A).	Write Intro (therapeutic motivation), Results (B) and Discussion (limitations).
Total	~10–11 h active work	~10–11 h active work

How the two tracks connect

- **Interface = `.h5ad` files.** Every trained model saves cell positions in `adata.obsm`, gene and protein positions in `adata.varm`, plus a `metrics.json`. Person B's analysis code runs on any such file, so it works on Person A's models without extra effort.
- **No waiting on each other:** each person trains on their own laptop.
- **Possible headline result:** *"A multimodal joint embedding makes drug-response programs readable and robust to technical artifacts"*. The opposite finding would also be worth reporting.

Deliverables

A 10-minute presentation (7 min talk + 3 min Q&A), a technical report, and a documented, reproducible code repository.

Main risks

- **Training time** on a laptop is unknown until we measure it on Saturday morning. If one run takes more than ~40 min, we shrink the grid (fewer cells, epochs or seeds).
- **sci-Plex is large.** We use a subset of cell lines and doses.